

Formal Verification: Content-Addressed Build Attestation

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Abstract

We formalize the content-addressed build attestation system using `uor.foundation` URIs. Every build artifact is content-addressed via its hash, and discharge proofs attest what happened during a build. Reproducible builds are those with no time or random access.

1 Introduction

The attestation system provides verifiable build provenance. Content addressing ensures that artifacts are uniquely identified by their content, independent of build time or location. This enables reproducible builds and supply chain verification.

2 Definitions

Definition 1 (StorePath). *Content-addressed store path: `uor.foundation/{hash}/{name}`.*

Definition 2 (Derivation). *Build recipe: inputs, builder, arguments.*

Definition 3 (DischargeProof). *Build evidence: derivation hash, output hashes, builder, timing, access flags.*

3 Theorems and Axioms

Axiom 4 (`hash_injective`). *Content hash is injective: same hash implies same content (random oracle model).*

Theorem 5 (`reproducible_no_time`). *Reproducible proofs have no time access.*

Proof. Decompose the Boolean conjunction. □

Theorem 6 (`reproducible_no_random`). *Reproducible proofs have no random access.*

Theorem 7 (`content_determines_path`). *Same content yields same hash yields same store path.*

Proof. Direct application of `hash_injective`. □

Theorem 8 (`same_path_same_uri`). *Two store paths with the same hash and name produce the same URI.*

4 Lean Source

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  Content-Addressed Build Attestation

  Every build artifact is content-addressed via uor.foundation URIs.
  Discharge proofs attest what happened during a build.
  Signatures bind ALL security-relevant fields.

  Address scheme: uor.foundation/{hash}/{name}

  Maps to:
  - hanzo/attest: attestation service (Go)
  - hanzo/kms: signing keys
  - lux/mpc: threshold signing for attestations
  - lux/consensus: on-chain attestation recording

  Properties:
  - Content addressing is injective (modulo hash axiom)
  - Attestation signatures are unforgeable
  - Reproducible builds have no time/random accesses
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import Mathlib.Data.Nat.Defs
import Mathlib.Tactic

namespace Build.Attestation

/-- Content-addressed store path: uor.foundation/{hash}/{name} -/
structure StorePath where
  hash : Nat          -- SHA-256 of content
  name : String       -- Human-readable name
  deriving DecidableEq, Repr

/-- Hash function (axiomatized) -/
axiom contentHash : Nat → Nat

/-- Random oracle: hash is injective (strongest axiom) -/
axiom hash_injective : a b : Nat, contentHash a = contentHash b → a = b

/-- Build derivation: recipe for producing artifacts -/
structure Derivation where
  inputs : List StorePath
  builder : StorePath
  args : List String
  deriving DecidableEq, Repr

/-- Derivation hash covers ALL fields -/
axiom derivationHash : Derivation → Nat

/-- Discharge proof: evidence of what a build did -/
structure DischargeProof where
  derivation : Nat          -- hash of derivation
  outputHashes : List Nat  -- hashes of outputs
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builder : Nat          -- builder's public key
startTime : Nat        -- build start
endTime : Nat          -- build end
hasTimeAccess : Bool   -- did it read the clock?
hasRandomAccess : Bool -- did it use entropy?

/-- Proof message binds ALL fields (prevents replay) -/
def proofMessage (p : DischargeProof) : Nat :=
  -- In practice: hash of (derivation, outputs, builder, times)
  p.derivation + p.outputHashes.length + p.builder + p.startTime + p.endTime

/-- Signature verification (axiomatized) -/
axiom sigVerify : Nat → Nat → Nat → Bool

/-- A proof is reproducible iff no time or random access -/
def DischargeProof.isReproducible (p : DischargeProof) : Bool :=
  !p.hasTimeAccess && !p.hasRandomAccess

/-- Reproducible proofs have no time access -/
theorem reproducible_no_time (p : DischargeProof)
  (h : p.isReproducible = true) :
  p.hasTimeAccess = false := by
  simp [DischargeProof.isReproducible, Bool.and_eq_true, Bool.not_eq_true'] at
    h
  exact h.1

/-- Reproducible proofs have no random access -/
theorem reproducible_no_random (p : DischargeProof)
  (h : p.isReproducible = true) :
  p.hasRandomAccess = false := by
  simp [DischargeProof.isReproducible, Bool.and_eq_true, Bool.not_eq_true'] at
    h
  exact h.2

/-- Same content → same hash → same store path -/
theorem content_determines_path (a b : Nat)
  (h : contentHash a = contentHash b) : a = b :=
  hash_injective a b h

/-- uor.foundation URI construction -/
def toURI (sp : StorePath) : String :=
  s!"uor.foundation/{sp.hash}/{sp.name}"

/-- Two store paths with same hash and name produce same URI -/
theorem same_path_same_uri (a b : StorePath)
  (hh : a.hash = b.hash) (hn : a.name = b.name) :
  toURI a = toURI b := by
  simp [toURI, hh, hn]

end Build.Attestation

```

5 References

Source code: <https://github.com/luxfi/proofs/blob/main/lean/Build/Attestation.lean>

White papers: <https://papers.lux.network>